

Decoding India's Western Export Landscape

A Predictive Analytics Approach to Identify Emerging Trends

B. Veda Vishnu Datta

Setting the Stage

- **India's trade with Western nations has been pivotal** in shaping its global economic presence.
- **Western countries account for a major share of India's high-value exports**, making them crucial for economic forecasting and policy planning.
- *"What can India's export data to the West reveal about evolving trade patterns?"*
- The goal became to **explore hidden patterns, identify top-performing commodities, and analyse regional trends.**
- Ultimately, this analysis aims to support **data-driven export strategies** for policymakers, businesses, and researchers.

Color Pallete

Color Name	Hex Code	Usage
Indigo Blue	#3F51B5	Slide titles, chart lines
Teal Green	#009688	Bar charts, visual comparisons
Golden Yellow	#FFD700	Top values, callouts
Cool Gray	#607D8B	Chart gridlines, footnotes
Off-White	#FAFAFA	Background
Charcoal Black	#212121	Main text

The Challenge

- Identify **top export commodities** across Western nations
- Analyse **country-wise trade performance** over time
- Examine **yearly** and **seasonal trends** in export volume and value
- Compare **quantity vs. value** to uncover high-value commodities
- Support **data-driven decision-making** for trade strategy



Meet the Data

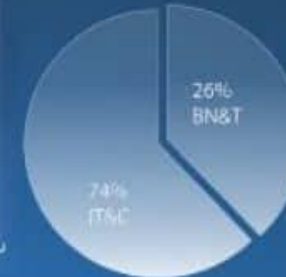
id	date	country_name	alpha_3_code	country_code	region	region_code	sub_region	sub_region_code	hs_code	commodity	unit	value_qt	value_rs	value_dl	year	month
2536484	01-10-2015	Belgium	BEL	56	Europe	150	Western Europe	155	61045990	Skirts Of Other Fibres	Nos	5.5	10.74	0.02	2015	10
1336247	01-03-2017	Kingdom of the Netherlands	NLD	528	Europe	150	Western Europe	155	63090000	Worn Clothing And Other Worn Articles	Kgs	5.75	20.44	0.03	2017	3
681447	01-01-2021	Mexico	MEX	484	Americas	19	Latin America and the Caribbean	419	44191900	Other	Kgs	0.32	1.73	0.020233918128654972	2021	1
899552	01-12-2023	France	FRA	250	Europe	150	Western Europe	155	63049241	Table Cloth And Table Covers Of Handloom	Nos	11.97	45.99	0.06	2023	12
1556908	01-09-2021	United States of America	USA	840	Americas	19	Northern America	21	92021000	Strng Musical Instruments Played Wth A Bow	Nos	0.29	1.75	0.02046783625730994	2021	9

Cleaning the Canvas

- Removed **duplicates** and handled **missing values**
- Converted **date** to **datetime** type
- Extracted **year** and **month** from the date column
- Filtered only relevant **Western countries**/regions
- Verified **data types** and **null values**



Uncovering the Trends



Distribution of market share among the major industry players: IT & C and BN & T was 74% and 26% percent respectively. A further change in the economic situation in the market will be characterized by a more equal distribution of market share major players.

Share of market activity



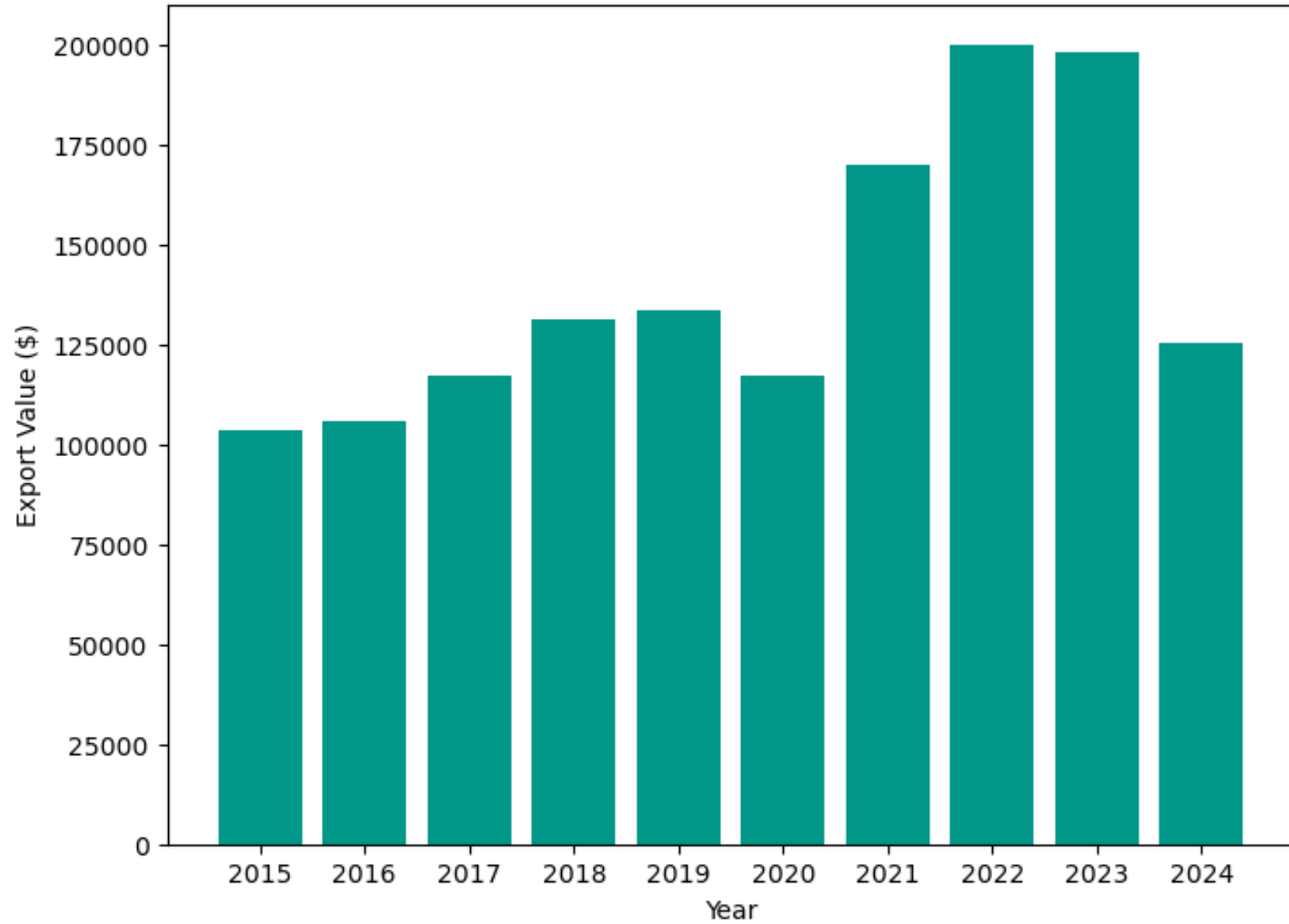
Changes in the activity of the active and passive market is uncertain. Established positive trends in various market segments.

Projected sales of main products in 2013

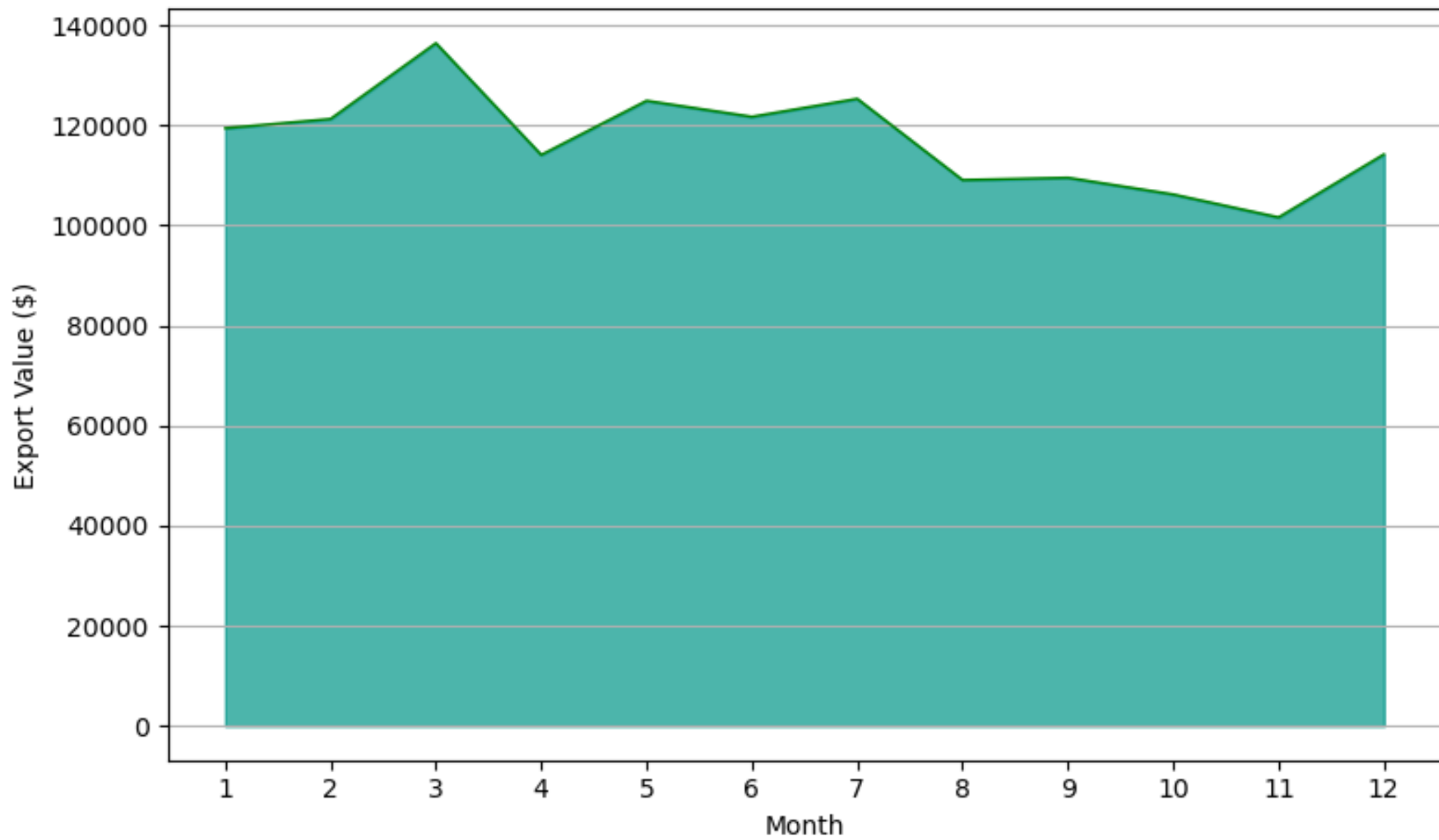


Passive market share

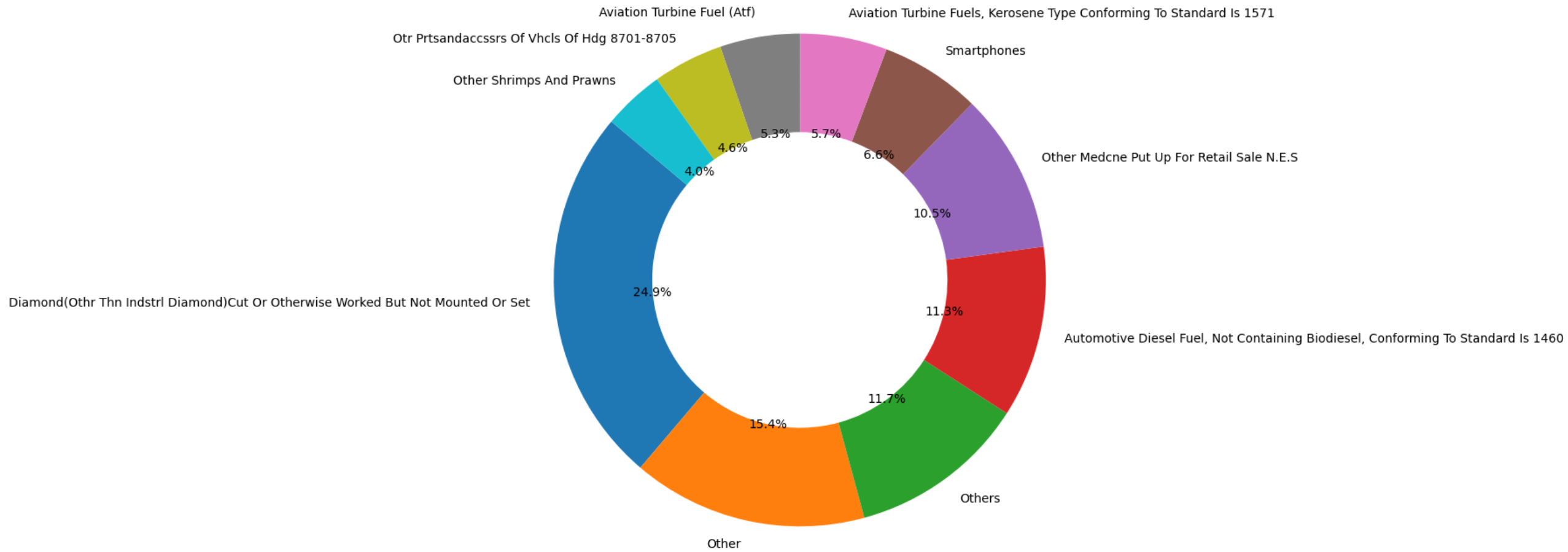
Total Export Value by Year (\$)



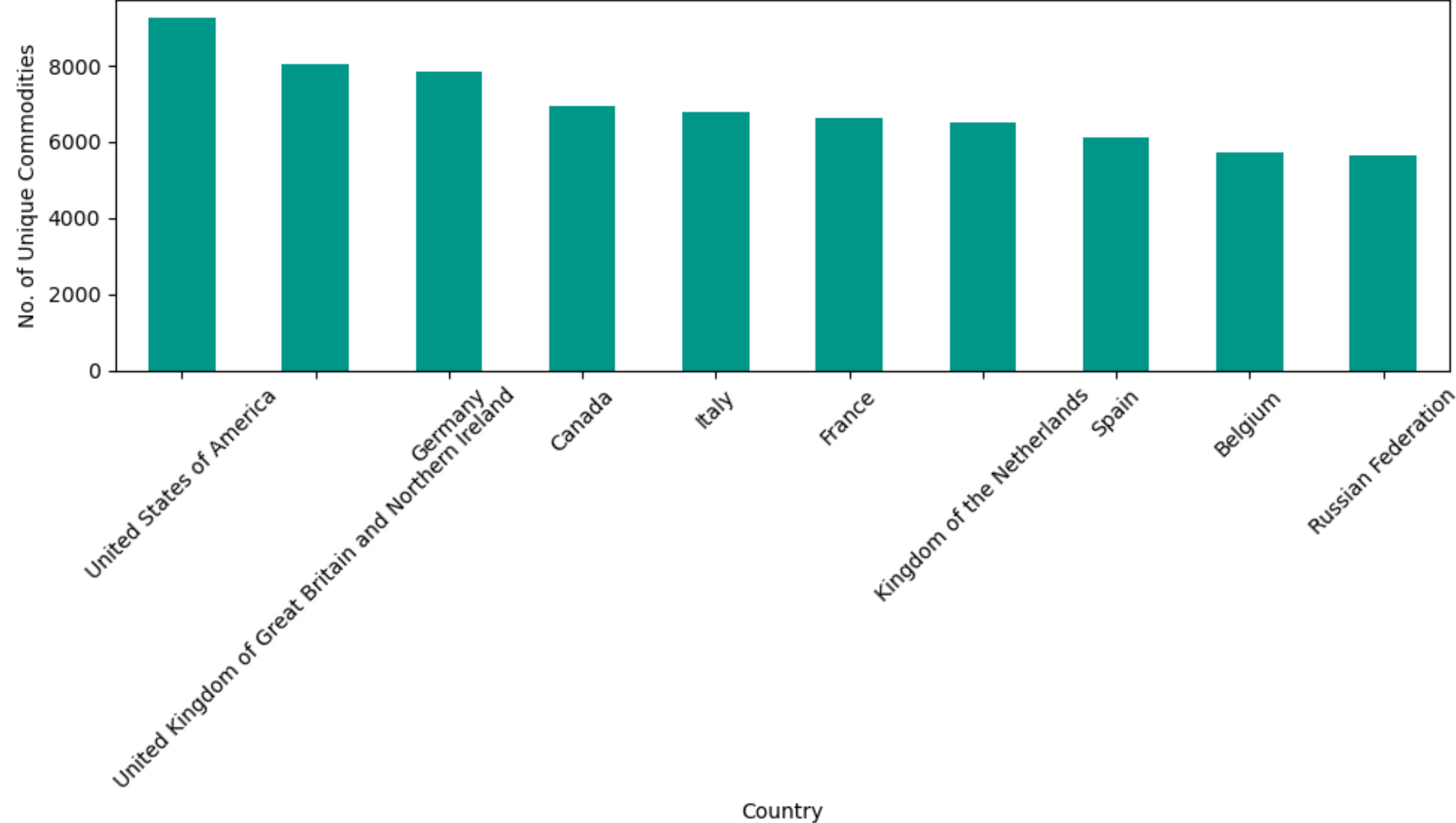
Monthly Export Value



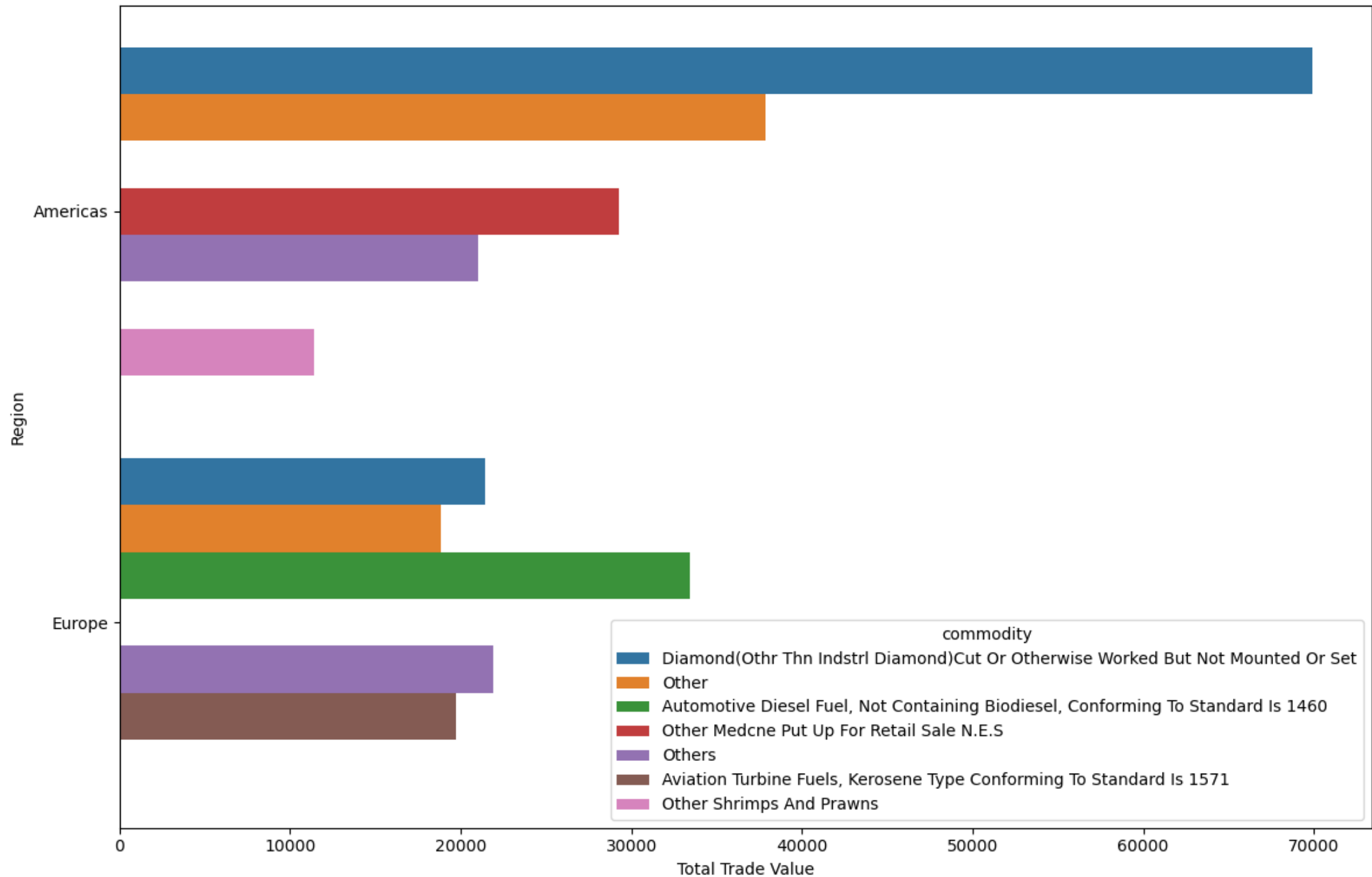
Top 10 Commodities Exported (usd)



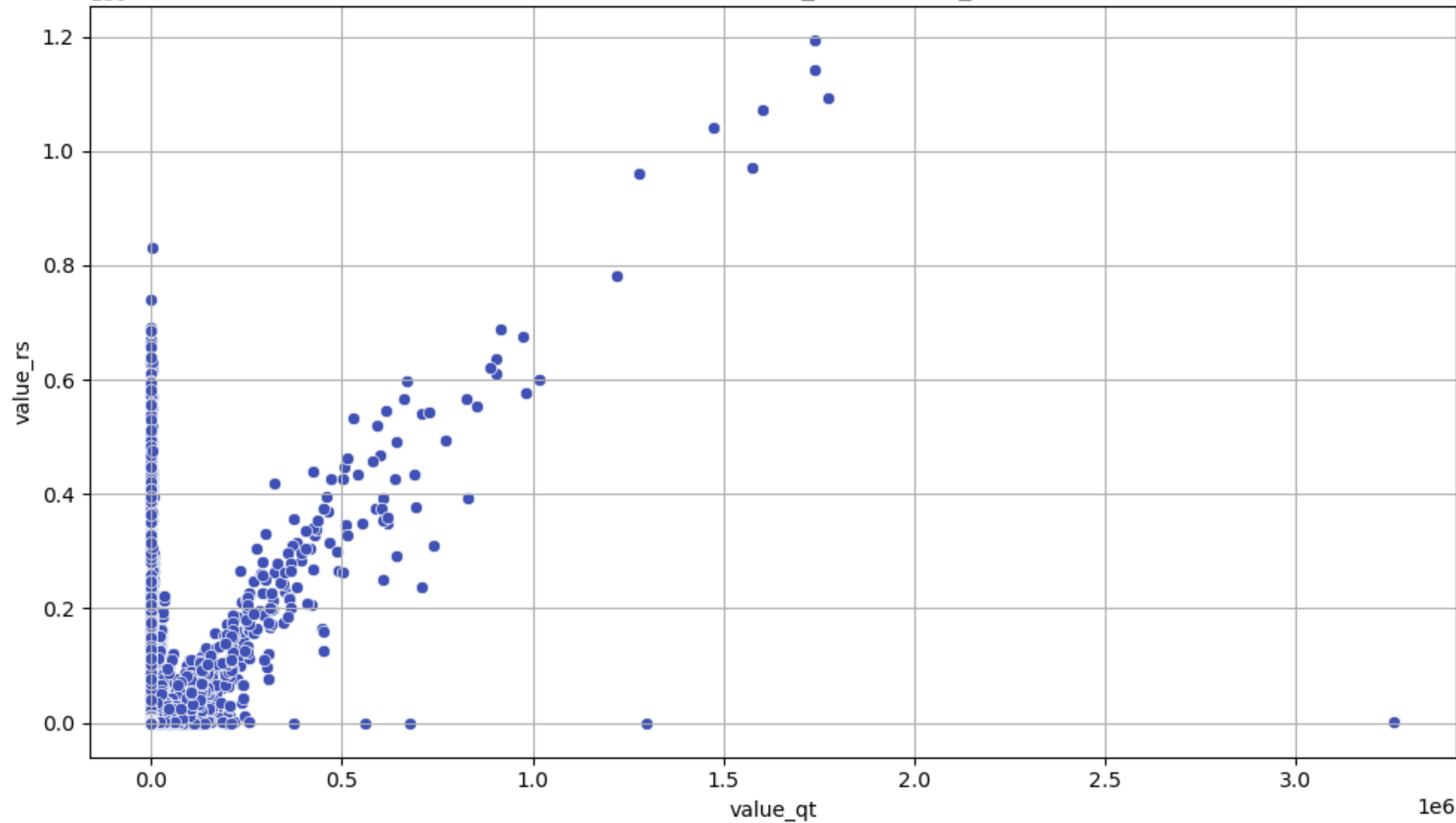
Top 10 Countries by Commodity Diversity



Top 5 Commodities by Total Value in Each Region



1e6



Year	New Trading Partners
2016	None
2017	Jersey, Guernsey, Saint Pierre and Miquelon, Holy See (Vatican City), Greenland, Sint Maarten (Dutch part)
2018	Falkland Islands (Malvinas)
2019	Sint Maarten (Dutch part)
2020	Svalbard and Jan Mayen
2021	Falkland Islands (Malvinas)
2022	Sint Maarten (Dutch part), Czechia
2023	Svalbard and Jan Mayen, Falkland Islands (Malvinas)
2024	Guernsey

Country Name	Commodity	Share
Falkland Islands (Malvinas)	Other Garments Of Cotton	92.24
Faroe Islands	Made Up Fishing Nets Othr Than Nylon	81.61
Guernsey	Jewellery Of Gold Set With Diamond	96.16
Jersey	Sim Cards	100.00
Sint Maarten (Dutch part)	Of Gold, Studded With Diamonds Of Heading 7102	87.49
Svalbard and Jan Mayen	Rice Parboiled	98.91

Vision for the Future

- **Find new markets** like Africa and South America.
- **Grow tech exports** like pharma and electronics.
- **Make better trade deals** with other countries.
- **Improve transport and shipping systems.**
- **Support small businesses** to export more.
- **Focus on eco-friendly exports.**



What It All Means

- **Exports are growing steadily** over the years.
- **Petroleum, jewellery, and Agri - products** lead exports.
- **Top markets** include USA, UAE, and China.
- **Shift towards tech-based exports** like pharma and electronics.
- **New markets** in Africa and Latin America offer potential.
- **Govt. policies** are boosting export performance.
- **Challenges** include global slowdown and regulations.



Curious Minds Welcome



Thank You